

Experiment 2: Synthesis of a commercial polymer, Nylon 6,6

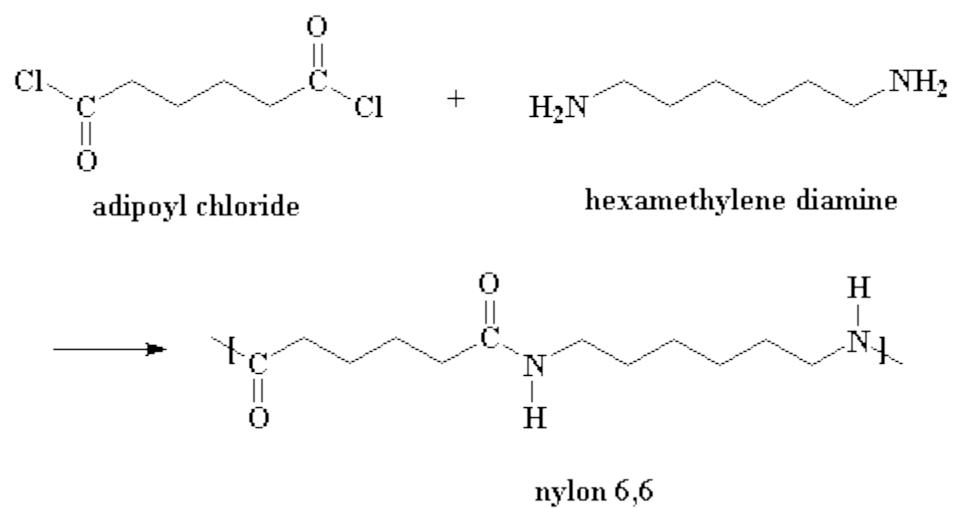
Introduction

The two most important kinds of nylon are nylon 6,6 and nylon 6. These two nylons have almost identical properties. Both were invented in the late 1930s. Nylon 6,6 was discovered first. It was invented in the United States by Wallace Carothers who was working for DuPont. Nylon 6,6 is a polyamide based polymeric material. It is commonly used in textile and plastic industries. Nylon 6,6 is normally produced by polycondensation of hexamethylenediamine and adipoyl chloride or adipic acid. It is a step-growth polymerization that produces high molecular weight Nylon 6,6.

Materials: Adipoyl chloride, Hexamethylenediamine, NaOH, Cyclohexane

Experimental Procedure:

1. First, measure the FTIR spectra of the pure starting materials, i.e., Adipoyl chloride and Hexamethylenediamine
2. Dissolve adipoyl chloride in Cyclohexane (1% v/v).
3. Add hexamethylenediamine to the 0.5M NaOH solution (1% v/v) (for polymerization under basic condition).
4. Add hexamethylenediamine to water (1% v/v) (for polymerization under neutral condition).
5. Add 5 ml of hexamethylenediamine solutions to two watch glasses or Petri dishes.
6. Add Adipoyl Chloride solution drop by drop on both the solution surfaces obtained in step 3 and step 4.
7. Let the polymerization reaction occur, forming Nylon 6,6 and releasing hydrogen chloride (HCl) as a byproduct.
8. Lift the polymer formed with forceps from the solvent.
9. Drain the excess solution and collect in an Eppendorf tube.
10. Wash the resulting polymer with water, dry it first with some tissue paper, and then with a heat gun (gentle heating).
11. Take the weight of the resulting polymer
12. Measure FTIR spectra of the resulting polymers and compare with the IR spectra of starting materials.



Scheme 1. Synthesis of Nylon 6,6 from adipoyl chloride and hexamethylene diamine.